Trailer Controller Pinout Reference

Hardware Architecture: Arduino Mega 2560 | Source Base: include/pins.h

1. System Indicators, Power & Communication Bus

Pin / Bus Connection	Variable / Function Label	I/O Classification	Hardware Integration Detail
5V	SYSTEM VCC (+5V)	Power Output	Logic supply rail for OLED screen and sensors
GND	SYSTEM GND	Ground Bus	Common ground reference for circuitry & display
Pin 13	STATUS_LED	Digital Output	System status heartbeat indicator
Pin 20	SDA	I2C Data Bus	Hardware I2C Data line for SSD1309 OLED
Pin 21	SCL	I2C Clock Bus	Hardware I2C Clock line for SSD1309 OLED

2. Input Control Switch Matrix

Arduino Pin	Code Constant Identifier	I/O Classification	Hardware Functional Description
Pin 22	L1_SW_A	Digital Input	Light 1 Toggle Switch Location A
Pin 23	L1_SW_B	Digital Input	Light 1 Toggle Switch Location B
Pin 24	L2_SW_A	Digital Input	Light 2 Toggle Switch Location A
Pin 25	L2_SW_B	Digital Input	Light 2 Toggle Switch Location B
Pin 26	L3_SW_A	Digital Input	Light 3 Toggle Switch Location A
Pin 27	L3_SW_B	Digital Input	Light 3 Toggle Switch Location B
Pin 28	L4_SW_A	Digital Input	Light 4 Toggle Switch Location A
Pin 29	L4_SW_B	Digital Input	Light 4 Toggle Switch Location B
Pin 30	LED_STRIP_SW	Digital Input	Dedicated LED Strip Control Switch
Pin 31	POD_LIGHT_SW	Digital Input	Dedicated Pod Light Control Switch
Pin 32	WINCH_UP_SW	Digital Input	Winch Actuation UP Button Input
Pin 33	WINCH_DN_SW	Digital Input	Winch Actuation DOWN Button Input

3. Power Actuation Channel Outputs

Arduino Pin	Code Constant Identifier	I/O Classification	Hardware Driver Target
Pin 34	LIGHT1_OUT	Digital Output	Light Circuit 1 Gate Driver / MOSFET
Pin 35	LIGHT2_OUT	Digital Output	Light Circuit 2 Gate Driver / MOSFET
Pin 36	LIGHT3_OUT	Digital Output	Light Circuit 3 Gate Driver / MOSFET
Pin 37	LIGHT4_OUT	Digital Output	Light Circuit 4 Gate Driver / MOSFET
Pin 38	LED_STRIP_OUT	Digital Output	LED Strip Gate Driver / MOSFET Out
Pin 39	POD_LIGHT_OUT	Digital Output	Pod Light Gate Driver / MOSFET Out
Pin 40	WINCH_UP_OUT	Digital Output	Winch UP Relay Contactor (Interlocked)
Pin 41	WINCH_DN_OUT	Digital Output	Winch DOWN Relay Contactor (Interlocked)
Pin 42	TUBIVATOR_OUT	Digital Output	Auxiliary Channel Output Driver